



1. Description

1.1. Project

Project Name	RAK3172-SiP-FreeRTOS
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	02/01/2026

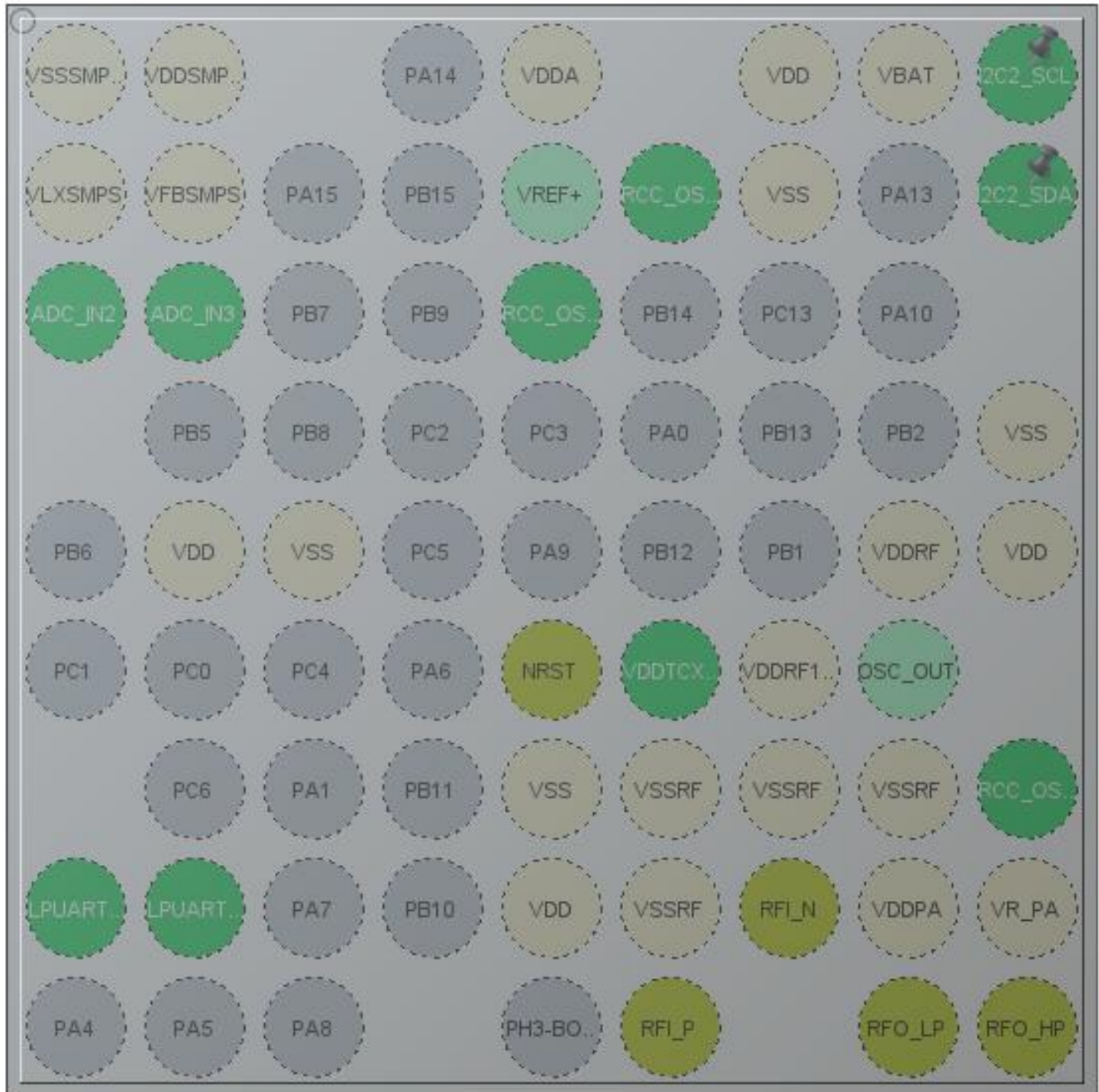
1.2. MCU

MCU Series	STM32WL
MCU Line	STM32WLEx
MCU name	STM32WLE5JCIx
MCU Package	UFBGA73
MCU Pin number	73

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



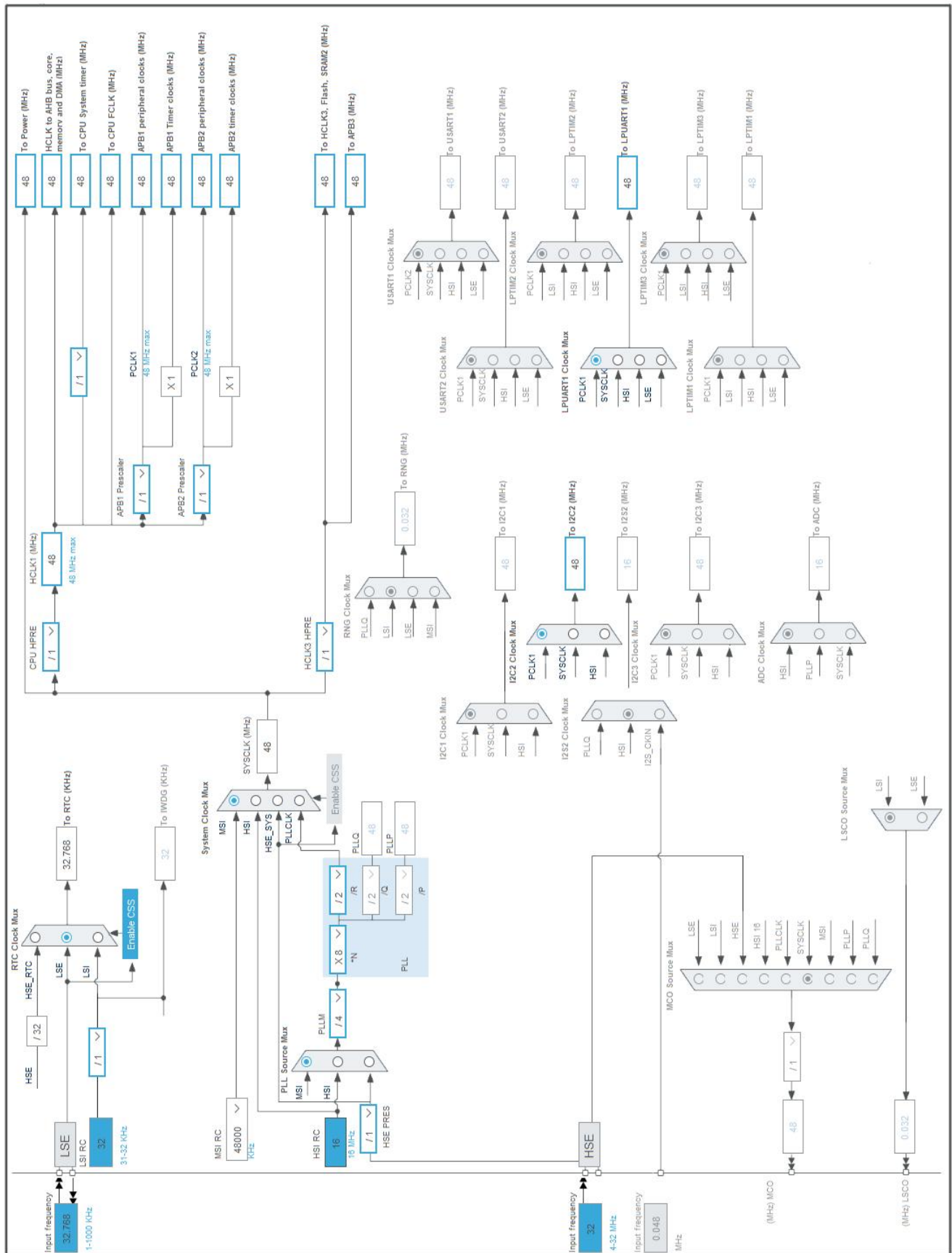
UFBGA73 (Top view)

3. Pins Configuration

Pin Number UFBGA73	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSSSMPS	Power		
A2	VDDSMPS	Power		
A5	VDDA	Power		
A7	VDD	Power		
A8	VBAT	Power		
A9	PA12	I/O	I2C2_SCL	
B1	VLXSMPS	Power		
B2	VFBSMPS	Power		
B6	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
B7	VSS	Power		
B9	PA11	I/O	I2C2_SDA	
C1	PB3	I/O	ADC_IN2	
C2	PB4	I/O	ADC_IN3	
C5	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
D9	VSS	Power		
E2	VDD	Power		
E3	VSS	Power		
E8	VDDRF	Power		
E9	VDD	Power		
F5	NRST	Reset		
F6	PB0-VDD_TCXO	I/O	VDDTCXO	
F7	VDDRF1V55	Power		
G5	VSS	Power		
G6	VSSRF	Power		
G7	VSSRF	Power		
G8	VSSRF	Power		
G9	OSC_IN	MonolO	RCC_OSC_IN	
H1	PA3	I/O	LPUART1_RX	
H2	PA2	I/O	LPUART1_TX	
H5	VDD	Power		
H6	VSSRF	Power		
H7	RFI_N	MonolO		
H8	VDDPA	Power		
H9	VR_PA	Power		
J6	RFI_P	MonolO		
J8	RFO_LP	MonolO		

Pin Number UFBGA73	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
J9	RFO_HP	MonolO		

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32WL
Line	STM32WLEx
MCU	STM32WLE5JCIx
Datasheet	DS13105 Rev7

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(AAA700)
Capacity	700.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	10.0 mA
Max Pulse Current	30.0 mA
Cells in series	1
Cells in parallel	1

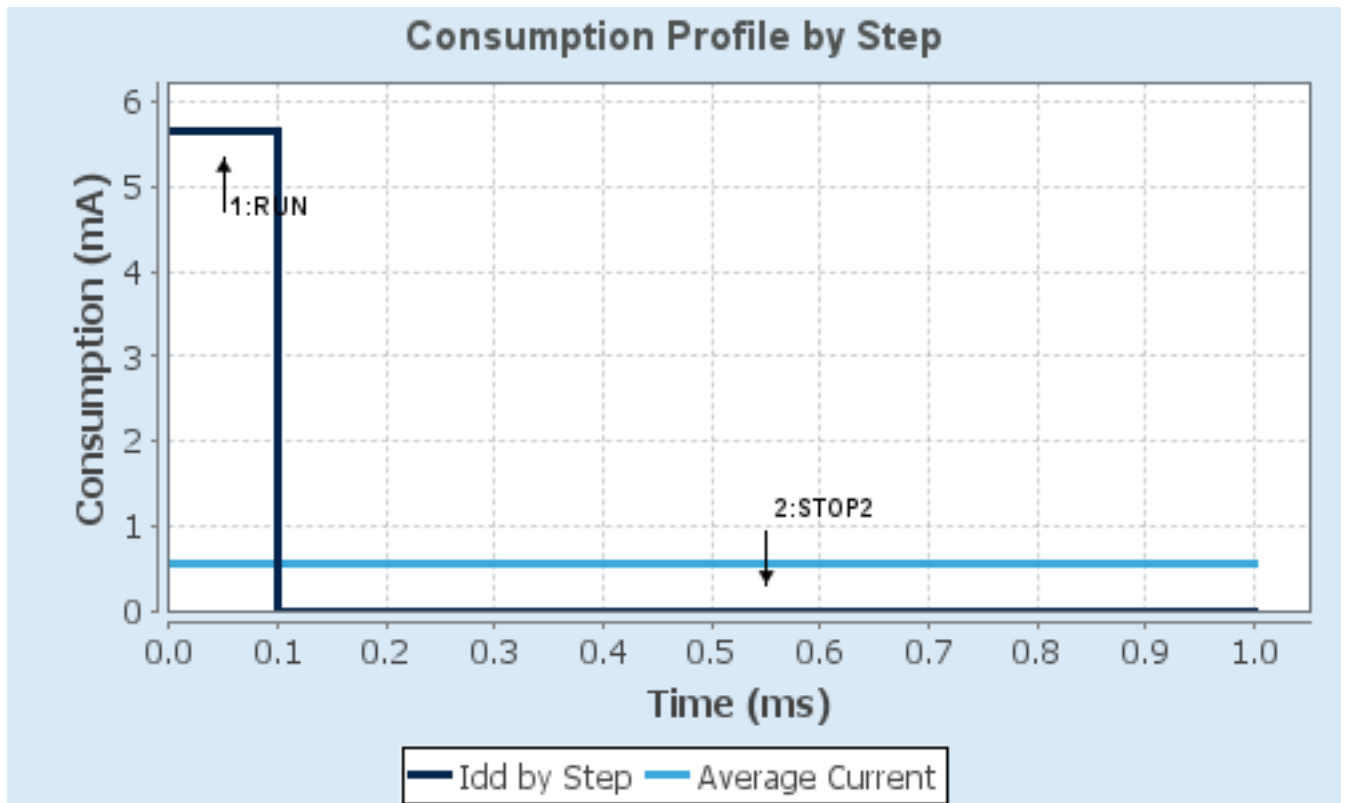
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP2
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Medium/SMPS-OFF	NoRange
Fetch Type	SRAM1	NA
CPU Frequency	48 MHz	0 Hz
Clock Configuration	MSI	ALL_CLOCKS_OFF
Clock Source Frequency	48 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	5.65 mA	885 nA
Duration	0.1 ms	0.9 ms
DMIPS	60.0	0.0
Ta Max	124.47	125
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	565.8 μ A
Battery Life	1 month, 21 days, 1 hour	Average DMIPS	60.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	RAK3172-SiP-FreeRTOS
Project Folder	C:\Users\danam\STM32CubeIDE\Alt-Workspace\RAK3172-SiP-FreeRTOS
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_WL V1.4.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_ADC_Init	ADC
5	MX_LPUART1_UART_Init	LPUART1
6	MX_RTC_Init	RTC
7	MX_SUBGHZ_Init	SUBGHZ
8	MX_LoRaWAN_Init	LORAWAN
9	MX_I2C2_Init	I2C2

3. *Peripherals and Middlewares Configuration*

3.1. ADC

mode: IN2

mode: IN3

mode: Temperature Sensor Channel

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADC_Settings:

Clock Prescaler	Synchronous clock mode divided by 2
Resolution	ADC 12-bit resolution
Calibration	Disable
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Auto Off	Enabled *
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data overwritten *
Sequencer	Sequencer set to fully configurable
SamplingTime Common 1	160.5 Cycles *
SamplingTime Common 2	160.5 Cycles *
Oversampling Mode	Enabled *
Right Bit Shift	8 bit shift *
Triggered Mode	Single trigger
Trigger Frequency	High frequency
Ratio	Oversampling ratio 256x *

ADC_Regular_ConversionMode:

Enable Regular Conversions	Disable
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3.2. ADV_TRACE

mode: Enabled

3.3. I2C2

I2C: I2C

3.3.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x10805D88 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.4. LPUART1

Mode: Asynchronous

3.4.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
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RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.5. MISC

mode: Enabled

3.6. RCC

High Speed Clock (HSE): TCXO

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.6.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	2 WS (3 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
MSI Calibration Value	0
MSI Auto Calibration	Enabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
LSE Drive Capability	LSE oscillator low drive capability

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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3.7. RTC

mode: Activate Clock Source

mode: Activate Calendar

Alarm A: Internal Alarm A

3.7.1. Parameter Settings:

General:

Asynchronous Predivider value

Bin Mode

SSRU Underflow Interrupt

Alarm A:

Free running 32 bit value

Binary AutoControl

Free running 32 bit mask

RTC_PREDIV_A *

Free running Binary mode *

Enabled

0

RTC_ALARMSUBSECONDBIN_AUTOCLR_NO *

SS[31:0] are compared and must match to activate alarm

3.8. SUBGHZ

mode: Activated

3.8.1. Parameter Settings:

Baudrate Prescaler Value

4 *

3.9. SYS

Timebase Source: None

3.10. TIMER

mode: Enabled

3.11. TINY_LPM

mode: Enabled

3.12. FREERTOS

Interface: CMSIS_V2

3.12.1. Config parameters:

API:

FreeRTOS API

CMSIS v2

Versions:

FreeRTOS version

10.6.2

CMSIS-RTOS version

2.1.3

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
USE_SB_COMPLETED_CALLBACK	0
USE_MINI_LIST_ITEM	1
MINIMAL_STACK_SIZE	512 *
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Disabled *
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Built in functionality enabled *
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	8192 *
HEAP_CLEAR_MEMORY_ON_FREE	0
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	1024

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.12.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Enabled *
vTaskSuspend	Enabled
xTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Disabled *
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled

uxTaskGetStackHighWaterMark2 Disabled

3.12.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT Disabled

Project settings (see parameter description first):

Use FW pack heap file Enabled

3.13. LORAWAN

mode: Enabled

3.13.1. LoRaWAN application:

Application selection:

ThreadX LORAWAN_APP_MEM_POOL_SIZE

6144

Application

Application configuration recommendations

End Node skeleton *

!! Please read carefully Information panel below!!

board settings:

Send Tx on Timer or Button Evt

TX_ON_TIMER

Probes Lines in Platform Settings

false

lora_app:

Active region

LORAMAC_REGION_US915 *

Transmition duty cycle

30000 *

Application user port

2

Switch class port

3

Default class

CLASS_A

Default handler message state

Unconfirmed message

Handler Adaptive Data Rate

On

Default activation type

OTAA

Force rejoin at each reboot

true

Default Tx output power

TX_POWER_5 *

Default Unicast ping slots periodicity

4

Default reponse timeout for class b and class c confirmed downlink frames in milli seconds.

8000

sys_conf:

Trace verbose level

VLEVEL_H *

Enable Application Logging

true

Disable Low Power Mode

false

Enable Sensor false

3.13.2. LoRaWAN commissioning:

Commissioning:

Public network true

Current network ID 0

se-identity:

LoRaWAN device EUI 00,00,00,00,00,00,00,00

App/Join EUI 01,01,01,01,01,01,01,01

Application key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C

Network key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C

Device Address 00,00,00,00

Network session key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C

Application session key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C

lorawan_conf:

Enable Key read access true

3.13.3. LoRaWAN middleware:

lorawan_conf:

Region(s) selection please select the desired region(s) in the list below

Region Asia freq: 923 false

Region Australia freq: 915 false

Region China freq: 470 false

Region China freq: 779 false

Region Europe freq: 433 false

Region Europe freq: 868 true

Region Korea freq: 920 false

Region India freq: 865 false

Region USA freq: 915 true

Region Russia freq: 864 false

Enable Hybrid mode false

Enable LoRaMAC ClassB false

Enable the context management storage **true ***

Select the LoRaWAN Link Layer specification version v1.0.4

Enable the additional LoRaWAN packages false

radio_conf:

Radio maximum wakeup time (in ms) 1

radio_board_if:

Select radio Driver User Board

TCXO support true

DCDC support true

Tx Rfo Config **CONF_RFO_HP ***

mw_log_conf:

Enable Middleware log true

3.13.4. Platform Settings:

RTC	RTC
USART	LPUART1

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC	PB3	ADC_IN2	Analog mode	No pull-up and no pull-down	n/a	
	PB4	ADC_IN3	Analog mode	No pull-up and no pull-down	n/a	
I2C2	PA12	I2C2_SCL	Alternate Function Open Drain	Pull-up *	High *	
	PA11	I2C2_SDA	Alternate Function Open Drain	Pull-up *	High *	
LPUART1	PA3	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA2	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PB0-VDD_TCXO	VDDTCXO	n/a	n/a	n/a	
	OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	

4.2. DMA configuration

DMA request	Stream	Direction	Priority
LPUART1_TX	DMA1_Channel1	Memory To Peripheral	Low

LPUART1_TX: DMA1_Channel1 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
RTC Tamper, RTC TimeStamp, LSECSS and RTC SSRU Interrupts	true	5	0
DMA1 Channel 1 Interrupt	true	5	0
LPUART1 Interrupt	true	5	0
RTC Alarms (A and B) Interrupt	true	5	0
SUBGHZ Radio Interrupt	true	5	0
PVD and PVM detector	unused		
FLASH (CFI) global Interrupt	unused		
RCC Interrupt	unused		
ADC Interrupt	unused		
I2C2 Event Interrupt	unused		
I2C2 Error Interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	false
RTC Tamper, RTC TimeStamp, LSECSS and RTC SSRU Interrupts	false	true	true
DMA1 Channel 1 Interrupt	false	true	true

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
LPUART1 Interrupt	false	true	true
RTC Alarms (A and B) Interrupt	false	true	true
SUBGHZ Radio Interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware									
FREERTOS  LORAWAN 									
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities
DMA 	ADC 	RTC 	I2C2 						ADV_TRACE 
GPIO 			LPUART1 						MISC 
HWIC 			SUBGHZ 						TIMER 
RCC 									TIHY_LPM 
SYS 									

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32wl_bsdl.zip
HW Models	https://www.st.com/resource/en/hw_model/stm32wl_reference_board_mb1720-xo_ios_shield.7z
IBIS models	https://www.st.com/resource/en/ibis_model/stm32wl_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32wl_svd.zip
System View Description	https://www.st.com/resource/en/svd/stm32wl-svd.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1720_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1789_hp_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1789_lp_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1791_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1792_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1842-bom.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1720_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1720_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_hp_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_hp_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_lp_bdp.zip

Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_lp_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1791_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1791_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1792_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1792_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1842-bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1842-manufacturing.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1720_schematics.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1789_hp_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1789_lp_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1791_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1792_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1842-schematic.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf

Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32cube-monitor-wireless-longrange_rfttest.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32wl5x-ex-product-line-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/beyond-the-wires-exploring-bluetooth-and-lorawan-connectivity.pdf
Flyers	https://www.st.com/resource/en/flyer/flnucleolrwan.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wl5ex.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-

timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5406-how-to-build-a-lora-application-with-stm32cubewl-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5409-stm32cube-mcu-package-examples-for-stm32wl-series-stmicroelectronics.pdf

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